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TruChit: A Blockchain-Based Trusted Chit Fund System with Creditworthiness Evaluation

Akhilesh Sharma[a], Preeti Chandrakar[a,*]

[a]*Department of Computer Science & Engineering, National Institute of Technology, Raipur, India*

Abstract

Chit fund is a peer-to-peer saving and borrowing scheme operated among trusted groups of people. It is a reliable source of funds in emergencies with no guarantor and low-interest rate. Despite their enduring benefits, chit funds face challenges related to trust, transparency, and security. At present, unscrupulous subscribers might join chit-funds, borrow money, and make payment defaults. This leads to disruptions in contribution cycle and affects overall functioning. Additionally, non-transparent record-keeping and transaction processes, hinder participants from verifying fund activities, creating a susceptible environment for fraud and malpractice. These challenges are an obstacle to the sustainability and trustworthiness of the chit-fund system. Therefore, to overcome aforementioned challenges, this paper proposes TruChit, a blockchain-based chit fund system with creditworthiness evaluation framework. It leverages the Adaptive Neuro-Fuzzy Inference System (ANFIS) to assess the creditworthiness of subscribers, which ensures the credibility and reliability of individual participants. Further, proposes permissioned blockchain-assisted chit-fund framework with role-based access control to instill trust, security, and transparency within chit-fund operations. Moreover, the efficacy of TruChit is evaluated by analyzing credit score dataset and achieves 93.5% accuracy, which is better than other approaches.

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Keywords: ANFIS (Adaptive Neuro-Fuzzy Inference System); Blockchain; Chit Fund; Credit-Score.

1. Introduction

Chit fund is a Rotating Savings and Credit Association (ROSCA). It is a revolving financial scheme where individuals come together to borrow and invest funds [1]. This scheme is popular among people who do not have access to the formal sector or are beleaguered by its insurmountable formalities. It offers community microfinance and peer-to-peer lending with less paperwork, without any guarantor, and with low-interest rates [2]. The chit funds are popular in many countries and found worldwide by different names, such as Lun-hui in China, Ko in Japan, Consorcio in Brazil, ROSCA in USA, etc. [3]. In India, chit funds are mainly governed by the chit fund Act 1982[4].

* Corresponding author. Tel.: +91-700-317-1408 .
E-mail address: preeti.pubnitrr@gmail.com

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The success of chit funds entirely relies on trust in the system because chits are governed by the local government rather than the central regulatory authority [5]. Currently, the paper-based administrative system suffers from a lack of regulation and monitoring. This benefits unscrupulous collateral lenders to run various fraudulent lucrative schemes. These dishonest investment schemes collapsed and led to significant financial losses to their investors [6]. These scams include Sharddha scam(frauded $6 billion approximately), Pearls group(Rs 51000 crore), Rose valley(Rs 17000 crore), PACL scams(Rs 41900 crore), and many more [7]. In addition, the lack of trust and transparency among the involved parties causes the fear of uncertainty in the chit-fund company. The non-transparent bidding process may result in rigging the auctioning process. Resulting in the subscriber who wins the bid doesn't receive money in a specific time frame.

On the other hand, chit funds involve individuals contributing a fixed amount at regular intervals. Through an auction, a member is chosen to receive a chit pool at a specific time interval [8]. In the current scenario, it is difficult to determine the payment history and creditworthiness of the chit fund members. Therefore, if a fraudulent member joins chits, causes various negative consequences for the other participants and the overall functioning of the chit fund [9]. A fraudulent member intentionally default on their monthly contributions or fail to pay the full amount during the chit circle. Further, they manipulate the bidding process and win the bid amount at lower bid, depriving other legitimate members to receive good return on their investment. Moreover, after winning bid amount they may disappear without completing the chit-circle [10]. These activities lead to financial losses for other legitimate members. Therefore, it is essential to determine the credibility and reliability of members joining the chit fund.

In this context, credit scoring is a prominent way to check the credibility and reliability of any person[11]. It is a measure through which a lender determines the creditworthiness of a person, and to accomplish this, ANFIS is an effective tool. It automatically generates fuzzy rules and membership functions from data through a learning process. ANFIS offers the benefit of adaptability, automatic parameter optimization, and effective modeling of non-linear relationships, making it a potentially more powerful tool for credit scoring[12]. Additionally, chit funds can be leveraged with blockchain technology to avoid scams and prevent the financial loss of middle-class people. The properties of blockchain, such as transparency, immutability, and decentralization [13] enhance trust, security and unlock value creation in the chit fund system. Moreover, it addresses many challenges that help to reduce friction, such as data breaches, data manipulation, and a trustless environment. Along with it increases operational efficiency and transparency. It also eliminates record-keeping vulnerabilities, streamlines the transaction, and makes it useful for processing complaints and grievance redressal.

Considering the above facts, this paper proposes a permissioned blockchain-enabled chit fund system that offers trust, transparency, and security to the existing system. In addition, ANFIS based creditworthiness evaluation model for participating members mitigates the chit defaults. Specifically, the paper makes the following contributions:

- This paper presents a three-layer architecture based on ANFIS and blockchain technology that ensures the reliability and credibility of chit fund system.
- An ANFIS-based credit score evaluation mechanism intended to prevent unscrupulous subscribers from joining the chit fund system.
- In order to mitigate the limitations of the existing system, this paper designed a transparent and secure framework by leveraging the concept of blockchain technology.
- The proposed solution is compared with different credit scoring classifiers, and it shows that the accuracy of the proposed scheme is comparatively higher, and more effectively predicts the trust level.

Henceforth, the paper is organized as follows: Section 2 briefly discusses the related work based on chit funds and creditworthiness evaluation system. Subsequently, the proposed system overview with layered architecture for TruChit is described in Section 3. The next Section 4 discusses the experimental setup, and performance evaluation. Finally, conclusion and future work are outlined in Section 5.

2. Related Work

Chit fund is a traditional financial system based on informal sharing and mutual trust. However, the lack of trust and transparency in the system causes scams and loss of people hard-earned money. A viable solution to this problem is to leverage blockchain in the chit fund system. Consequently, the success of any chit completely relies on trust. In

this respect, a credit score is a valuable tool to check the creditworthiness of its members. Considering these issues, this section describes research articles on blockchain-based chit fund systems and effective credit score evaluation.

For instance, [8] proposed a blockchain-based chit fund model that is secure, transparent, and efficient than traditional systems. It completely replaces the legacy paper-based system. However, it does not ensure the trustworthiness of chit funds, has fewer transactions per second, and lacks secure registration of the customers. In addition, ChitChain, a blockchain-based chit fund system proposed by [9]. It facilitates peer-to-peer lending and borrowing, promotes transparency among participants, and reduces transaction costs. However, no attention has been paid to implementation and customer trust issues. Further, ChitMonk [14] developed blockchain-based chit solution to provide transparency, decentralization and serve as a financial inclusion tool. They launched a subscriber-centric App to provide better services to subscribers. Similarly, Govt. of Telangana, India, furthered this idea by adopting T-Chits [5], a blockchain-based chit fund solution that fully automates the chit fund process. The system is completely cashless, and the entire chit activities are digital, verifiable, and manageable.

In [15] the authors designed a trust lending circle supported by blockchain and implemented ROSCA under the Ethereum network. Further, [16] developed Decentralized Application (DApp) for ROSCA that uses smart contracts for peer-to-peer lending. It supports all the functionality of a trusted lending circle. Additionally, BX Smart Labs solution [17] designed a blockchain-based app for saving circles. Through this, chit funds are implemented remotely, securely, and with a high degree of transparency.

On the other hand, to mitigate the chit fund frauds, credit scoring plays a prominent part in the financial system, and it is also an invaluable tool for predicting credit defaults. The credit score is a measure through which creditworthiness of a person [18] can be determined. A high credit score results in high credibility and reliability of a person. Although a lot of research calculated credit scores by applying fuzzy logic, AI, ML, and DL. In [19] presented a fuzzy logic-based credit scoring model for microfinance. Using FL (Fuzzy logic) they evaluate customer behavior to reduce loan defaults. Specifically, it utilizes the Mamdani fuzzy inference system, trapezoidal membership function, and centroid method for calculating credit scores. In [20] author considered multi-criteria attributes (economic, environmental, and social attributes) for creating a sustainable credit scoring model. To overcome the decision-making ambiguity, they integrated fuzzy set theory with fuzzy best-worst method. They also identify the engagement of potential borrowers in sustainable business.

In [11], the authors presented a hybrid model based on ANFIS and Recurrent Neural Network(RNN) to predict the credit score. They prevented the loan offering to suspicious customers. Further, the authors in [21] considered weighing and balancing effects in logistic regression for presenting a credit scoring model. Next, a credit evaluation model based on intuitionistic fuzzy sets is given by [22]. The experts aggregate the opinions by employing the intuitionistic fuzzy number and determine the degree between the attributes used for credit score evaluation.

In summary, few researchers adopted blockchain to address challenges faced by traditional chit fund systems, such as lack of transparency, security, and efficiency [5, 9, 8, 14, 15, 16, 17]. However, the existing approaches have failed to provide much attention to the trust factor of the chit fund participants. Moreover, few researchers used fuzzy logic to find credit score but in FL, fuzzy rules and membership functions need to be defined manually by domain experts [18, 19, 20, 22]. This process is time-consuming and requires domain experts. Therefore, research is required for efficient creditworthiness evaluation and secure the chit fund system to increase trust, participation, confidence and reduce potential risks of scams.

3. Proposed System

The development of a blockchain-based trusted chit fund system with creditworthiness evaluation integrating blockchain technology, smart contracts, and credit scoring mechanisms to enhance transparency, security, trust, and efficiency in chit fund operations. This section presents the system overview and layered architecture of the proposed system.

3.1. System Overview

The proposed model consists three entities namely (1) Regulator($\mathcal{R}$), (2) Chit fund operator(CF_{ID}), and (3) Customers(CU_{ID}). Table 1 outlines the notations used in the paper. Subsequently, a brief description of the entities are as follows.

- ***Regulator($\mathcal{R}$):*** The regulator ($\mathcal{R}$) is a government authority responsible for regulating and overseeing chit fund activities to monitor illicit activities and prevent fraudulent practices by the participating entities. Here, $\mathcal{R}$ plays an administrative role and is assumed to be a fully trusted party that can not be compromised.
- ***Chit fund operator(CF_{ID}):*** The chit fund operator (CF_{ID}) starts and manages the chit fund. It is responsible for regulatory approvals, enrolling participants, conducting auction processes, and ensuring timely disbursement of payouts to participants.
- ***Customer(CU_{ID}):*** The customer (CU_{ID}) joins the chit fund as a member and contributes a fixed amount regularly to the fund. It is the end beneficiary of the chit process.

Table 1: Description of symbolic notation used in this article

Notations	Descriptions
E_{ID}	Entity
$\mathcal{R}$	Regulator
CF_{ID}	Chit fund operator
CU_{ID}	Customer
$CS_{CU_{ID}}$	Credit score of customer CU_{ID}
$R_{CU_{ID}}$	Credit recommendation
Pu_{ID}	Public key of entity E_{ID}
Pr_{ID}	Private key of entity E_{ID}
SK_{ID}	Secret key of entity E_{ID}
$E(.)$	Encryption operation
$D(.)$	Decryption operation
$Cert_{ID}$	Digital certificate of entity E_{ID}
$\mathcal{W}_{ID}$	Digital wallet of entity E_{ID}
D_{tx}	Record of the entity
$\|\|$	Concatenation operation
Tx	Transaction consists of data d
T_s	Time stamp

3.2. Layered Architecture

The layered architectural design of the proposed system comprises three different layers: Data processing layer, Credit rating layer, and Blockchain-based chit fund layer, as shown in Fig. 1. In the proposed ecosystem, Regulator($\mathcal{R}$), Chit fund operator (CF_{ID}), and the Customer(CU_{ID}) are key entities. Moreover, the CF_{ID} is the company or group of persons, who are responsible for conducting and managing chits. It needs to register and initially authenticate with $\mathcal{R}$. CU_{ID} are the investors or end beneficiaries of chits.

3.2.1. Data Processing Layer

The main components of this layer are the trust attributes TA_i for determining the creditworthiness of CU_{ID}. It majorly includes Delay_from_due_date ($\mathcal{T}_d$), Num_of_Delayed_Payment ($\mathcal{N}_d$), Outstanding_Debt ($\mathcal{O}_d$), Credit_Utilization_Ratio ($\mathcal{CUR}$). These attributes are collected and analyzed based on credit history of the customer. These multi-criteria trust attributes are preprocessed by data cleaning, conversion, and reducing the number of features. The processed data are used as input to the next layer for enhancing the accuracy of the customer's creditworthiness evaluation.

3.2.2. Credit Rating Layer

This layer proposes a creditworthiness computing model based on ANFIS. It evaluates credit score ($CS_{CU_{ID}}$) value of CU_{ID} and allows chit operators to examine the creditworthiness of customers. In order to measure the credit score value, TruChit integrated neural learning ability with Takagi Sugeno Fuzzy Inference System(TSFIS) to get the optimized parameters. The proposed ANFIS-based model encompasses the following steps: (i) Fuzzification, (ii) Inference engine, (iii) Normalization, (iv) Defuzzification, and (v) Output generation, as shown in Fig. 2. Firstly, the crisp input values of attributes $A_i(\mathcal{X}) \in \{\mathcal{T}_d,\ \mathcal{N}_d,\ \mathcal{O}_d,\ \mathcal{CUR}\}$ collected from physical layer are inputted into the

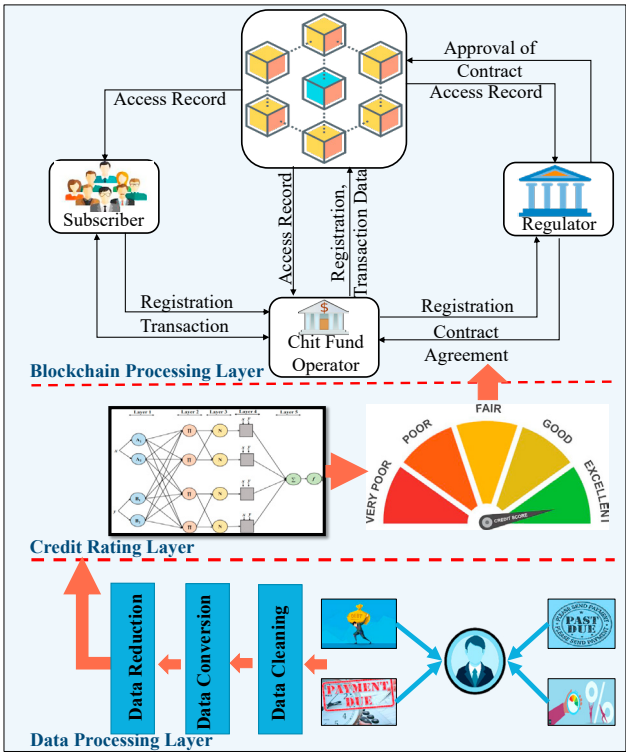

Fig. 1. System architecture of proposed model

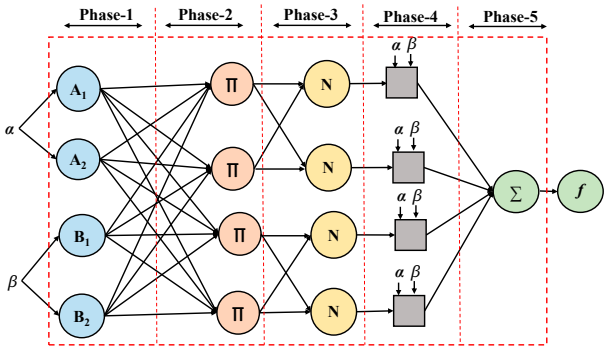

Fig. 2. Architecture of ANFIS

fuzzification process. Next, in the fuzzification process, based on membership function (μ), crisp values are converted into fuzzy values. The membership degrees $\mu(\mathcal{T}_d)$, $\mu(\mathcal{N}_d)$, $\mu(\mathcal{O}_d)$, and $\mu(\mathcal{CUR})$ quantify the membership functions for the input variables $\{\mathcal{T}_d, \mathcal{N}_d, \mathcal{O}_d, \mathcal{CUR}\}$ respectively. Particularly, the output of first phase O_i^1 in the ANFIS model is calculated using Eq. 1.

$$O_i^1 = \mu_{A_i}(\alpha) \qquad i = 1, 2, 3...n \tag{1}$$

The $\mu_{A_i}(\alpha)$ is the bell-shaped Gaussian membership function associated with input variable α is given by Eq. 2.

$$\mu_{A_i}(\alpha) = exp\left[-\frac{1}{2}\left|\frac{\alpha - c}{s}\right|^m \right] \tag{2}$$

Here, c and s represent the mean and standard deviation. The next phase is inference engine. The proposed model adopted Takagi Sugeno inference engine that follows if-then rules as:

$$if\ \alpha\ is\ A_i\ and\ \beta\ is\ B_i\ then\ z\ is\ f(\alpha, \beta)$$

Where α and β are the fuzzy set, and z is a crisp function. By using these membership degrees, the output of next phase is calculated using Eq. 3.

$$O_i^2 = \ddot{w}_i = \mu_{A_i}(\alpha) \times \mu_{B_i}(\beta) \qquad i = 1, 2, 3...n \tag{3}$$

Next, the rule aggregation phase computes normalized firing strength $\hat{w}^j$ of each jth rule. Here, $\hat{w}^j$ determines the contribution of jth rule to the overall output and is calculated by using Eq. 4.

$$O_i^3 = \hat{w}^j = \frac{\ddot{w}^j}{\Sigma_{i=1}^n \ddot{w}_i} \tag{4}$$

In the defuzzification phase, by using Eq. 5, the final output O_i^4 of each rule is calculated based on the firing strength $\hat{w}^j$ of the previous output O_i^3 and consequent parameter set (p, q, r).

$$O_i^4 = \hat{w}^j \times f_i = \hat{w}^j \times (p_i\alpha + q_i\beta + r_i) \qquad i = 1, 2, 3...n \tag{5}$$

Now in the last phase final output of the ANFIS model is computed by aggregating values from the previous output as given is Eq. 6.

$$CS_i = O_i^5 = \Sigma_{i=1}^n \hat{w}^j \times f_i = \frac{\Sigma_{i=1}^n \ddot{w}_i \times f_i}{\Sigma_{i=1}^n \ddot{w}_i} \qquad i = 1, 2, 3...n \tag{6}$$

The step-wise illustration of the credit score generation is given in Algorithm 1. Here, the algorithm provides a recommendation based on the calculated credit score CS_i such that $i \in CU_{ID}$ from Eq. 6. Subsequently, the customers are categorized into trustworthiness levels based on their credit scores.

Algorithm 1 Calculate credit score $CS_{CU_{ID}}$

Input:- Credit score attributes $\mathcal{T}_d$, $\mathcal{N}_d$, $\mathcal{O}_d$, and $\mathcal{CUR}$
output:- Customers credit recommendation $R_{CU_{ID}}$
Initialization:- *Initialize weight of layer* $i \longrightarrow \ddot{w}_i$
Input vector as $x_1, x_2, x_3, ..., x_n$ *from input layer*
Membership function for each fuzzy neuron in fuzzy layers:
$\ddot{w}_i = \mu_{A_i}(\alpha) \times \mu_{B_i}(\beta) \qquad i = 1, 2, 3...n$

1: Calculate the normalized firing strength of each rule from ANFIS model (using Eq. 4)
$\hat{w}^j = \frac{\ddot{w}^j}{\Sigma_{i=1}^n \ddot{w}_i}$
2: Calculate $CS_{CU_{ID}}$ credit score from ANFIS model (using Eq. 6)
$CS_{CU_{ID}} = \Sigma_{i=1}^n \hat{w}^j \times f_i = \frac{\Sigma_{i=1}^n \ddot{w}_i \times f_i}{\Sigma_{i=1}^n \ddot{w}_i} \qquad i = 1, 2, 3...n$
3: **if** ($CS_{CU_{ID}} < 60$) **then**
4: $R_{CU_{ID}}$="Non trustworthy"
5: **else**
6: $R_{CU_{ID}}$="Trustworthy"
7: **end if**

3.2.3. Blockchain-Based Chit Fund Layer

At the blockchain layer, the final computation value of credit score $CS_{CU_{ID}}$ with a time stamp (T_s) will be recorded in the blockchain, only after initial registration performed by the chit fund operator. Fig. 1 depicts the proposed blockchain-assisted creditworthiness computation model. In this layer the chit operators CF_{ID} and customers $CS_{CU_{ID}}$ compass different phases.

- ***Registration of Chit Fund Company:*** Each CF_{ID} participating in the permissioned blockchain network for chit fund systems follows registration processes to prevent unauthorized intervention. The process involves registering CF_{ID} with Regulator $\mathcal{R}$. CF_{ID} provides their identification information and other relevant details to $\mathcal{R}$. $\mathcal{R}$ verifies CF_{ID} identity and checks credit score. Once credit score verification is completed, the chit fund operator gets the digital certificate $Cert_{ID}$, which consists of (ID, Pu_{ID}), and a private key Pr_{ID} to sign transactions. Here, the Pr_{ID} is generated locally on the CF_{ID} for enhanced security. Simultaneously, a wallet $\mathcal{W}_{ID}$ is generated for each entity by computing $\mathcal{W}_{ID} = \{Cert_{ID}(ID, Pu_{ID})||Pr_{ID}||CS_{CF_{ID}}\}$. Then $\mathcal{R}$ creates CF_{ID} identity within the blockchain network also a digital wallet $\mathcal{W}_{ID} = \{Cert_{ID}(ID, Pu_{ID})||Pr_{ID}||CS_{CF_{ID}}\}$ generated for CF_{ID}.
- ***Registration of Customers:*** The registered CF_{ID} starts a chit and determines number of customers, chit amount, and duration. Thereafter, CF_{ID} registers CU_{ID} participating in chit fund follows the registration process to prevent fraud. CU_{ID} provides their identification information and other relevant details to CF_{ID}. Then CF_{ID} verifies CU_{ID} identity and checks credit score. The credit score is calculated through historical data and financial

behavior of CU_{ID}. After successfully evaluating of credit score, CU_{ID} are allowed to register in the chit. CU_{ID} selects any chit of his/her choice and starts the contribution. A wallet $\mathcal{W}_{ID} = \{Cert_{ID}(ID, Pu_{ID})||Pr_{ID}||CS_{CU_{ID}}\}$ also generated for CU_{ID}

- ***Access Control Phase:*** After the completion of registration phase, role-based access control(RBAC) policies and permissions for entities CF_{ID} and CU_{ID}'s are defined. RBAC determines privileges for CF_{ID} and CU_{ID} that they can perform operations within the blockchain network, based on their roles and responsibilities. The detailed description of roles and access control is illustrated in Table 2.

Table 2: Description of role-based access control

Entity	Access level	Permissions
$\mathcal{R}$	Admin	Read/write Endorsement Admin privileges, manage channel configuration, chit fund addition/removal
CF_{ID}	Member	Read/write Transaction endorsement
CU_{ID}	Restricted	Read only Initiate transaction

- ***Transaction Generation Phase:*** A transaction data(D_{tx}) containing chit, bidding or auctioning data is generated by $E_{ID} \in (CF_{ID}, CU_{ID})$. E_{ID} performs E(.) on D_{tx} by using SK_{ID} and computes C_{Tx}=$E_{SK_{ID}}(D_{tx})$. Meanwhile, a transaction record Tx is generated that will be stored in the blockchain. It consists of $Tx = ID \parallel T_s \parallel C_{Tx}$. The stepwise illustration of user registration and block creation is depicted in Algorithm 2.

Algorithm 2 User registration and block generation

Input:- Customers credit recommendation $R_{CU_{ID}}$
output:- A block added to blockchain

if ($R_{CU_{ID}} \in \{Trustworthy\}$) **then**
  $\mathcal{W}_{ID} = \{Cert_{ID}(ID, Pu_{ID})||Pr_{ID}||CS_{CU_{ID}}\}$.
  Apply role-based access control policies and permissions
  Membership Service Provider (MSP) assigned roles, rules, and permissions
  Generate transaction $Tx_{D_{tx}} = ID \parallel T_s \parallel C_{Tx}$
  $Transaction_Tx_{D_{tx}} \longleftarrow Endorse_Transaction(Tx_{D_{tx}})$ by endorsing peers
  $CreateBlock_Tx_{D_{tx}} \longleftarrow Validate(Tx_{D_{tx}}))$ by ordering peers
  $Add_BC_{CN} \longleftarrow CreateBlock_Tx_{D_{tx}}$ by committing peers
  $Ledger_Update(Tx_{D_{tx}}) \longleftarrow Transaction_Tx_{D_{tx}}$
else
  The customer is not trustworthy
end if

- ***Transaction Retrieval Phase:*** Whenever any $E'_{ID} \in (CF_{ID}, CU_{ID})$ tries to access the data from blockchain, the data owner entity provides access to its data by sharing SK_{ID}. The data owner entity computes $E_{Pu_{E'_{ID}}}(SK_{ID})$ and sends it to requester entity E'_{ID}. Subsequently, E'_{ID} computes $D_{Pr_{E'_{ID}}}(E_{Pu_{E'_{ID}}}(SK_{ID}))$. Once the E'_{ID} gets SK_{ID}, it will search the blockchain for the record Tx. Once Tx is fetched, E'_{ID} performs $D_{SK_{ID}}(D_{tx})$ and successfully retrieves the record D_{tx}.

4. Performance Analysis

This section presents the experimental setup and performance analysis of TruChit across two significant dimensions: (1) Performance measurement of the ANFIS model for credit score evaluation and (2) Execution time incurred during customer registration in the blockchain network.

4.1. Experimental Setup

In order to assess the feasibility and effectiveness of TruChit, the prototype is implemented on 3 nodes with Intel core i7 2.6 GHz processor, 16 GB of RAM, and 1 TB SSD under Ubuntu 22.02 LTS environment. Specifically, for data processing and credit rating layer, extensive experiments are conducted with the ANFIS model that leverages Takagi-Sugeno fuzzy inference system and Gaussian membership function. It utilizes credit score dataset [23], categorized into training data(80%) and testing data(20%). Further, the training is performed with 100 epochs to obtain minimum error between the desired output and actual output. The complete model is embedded with Hyperledger Fabric 2.2.3 based permissioned blockchain framework with performance benchmarking tool using Hyperledger Caliper 0.4.2.

4.2. Performance Measurement of ANFIS

The proposed system is tested on the credit score dataset [23] containing 10 features and 1000 instances. Subsequently, through the ANOVA feature extraction algorithm, four key features are extracted, namely $\{\mathcal{T}_d,\ \mathcal{N}_d,\ \mathcal{O}_d,\ \mathcal{CUR}\}$. The simulation of the ANFIS model calculates a numeric value for $CS_{CU_{ID}}$ based on input features and classifies as "Non trustworthy," or "Trustworthy". The foremost observation using the ANFIS model concerns the RMSE(Root Mean Square Error) variance across epochs, illustrated in Fig. 3. The RMSE value represents the error between the ANFIS model's predictions and the actual credit scores for the training data. It also provides insights into ANFIS model in learning and improving over time during the training process. From Fig. 3, it observes that initially, RMSE decreases rapidly as the model quickly learns the underlying patterns in the data. The RMSE error plateaued after 80 epochs, suggesting that the model has converged.

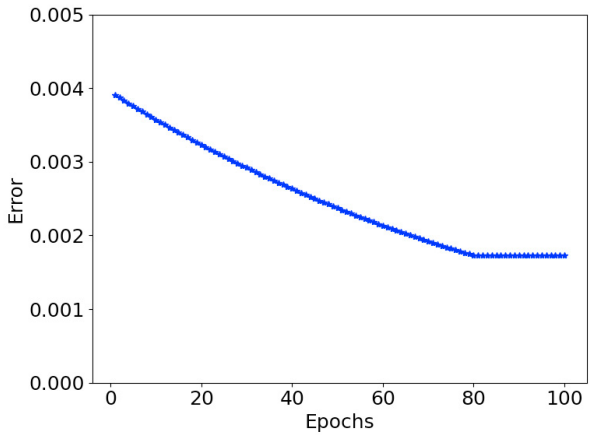

Fig. 3. Variation of root mean square error vs epochs

0 0.0% (TN)	**13** 6.5% (FP)	0.0% 100%
0 0.0% (FN)	**187** 93.50% (TP)	100% 0.0%
NaN% NaN%	93.50% 6.5%	93.50% 6.5%

Fig. 4. Confusion matrix of ANFIS classifier

Another performance metric is the confusion matrix depicted in Fig. 4 created on testing data. This matrix provides a comprehensive overview of the model's ability to classify individuals as trustworthy or non-trustworthy. Here, symbols are signified as TP: actually trustworthy and correctly classified as trustworthy, TN: actually non-trustworthy and correctly classified as non-trustworthy, FP: actually non-trustworthy but incorrectly classified as trustworthy, and FN: actually trustworthy but incorrectly classified as non-trustworthy. Subsequently, utilizing the confusion matrix with the testing data, the accuracy of the proposed model is calculated through the Eq. 7.

$$\sigma = \frac{TP + TN}{TP + FP + FN + TN} \tag{7}$$

Here, the accuracy(σ) is calculated as 93.50% by dividing correct predictions by the total number of predictions. It signifies the model's ability to make accurate creditworthiness classifications.

The performance of the ANFIS-based credit score evaluation is compared to other traditional credit scoring models, including Support Vector Machine (SVM), Naive Bayes, Decision Tree, Neural Network, and K-Nearest Neighbor

(KNN) classifiers. This comparison shown in Fig. 5 determines the accuracy of proposed solution is 93.5% and performs better as compared to other classifiers.

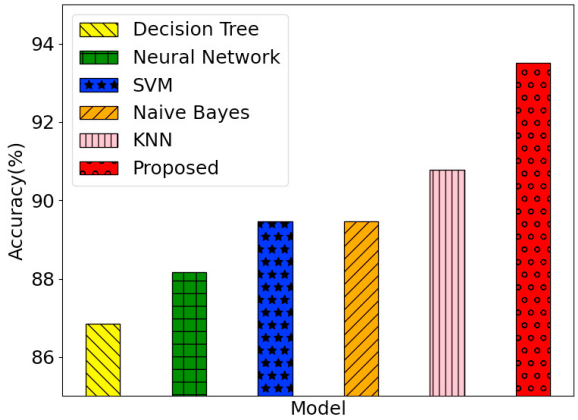

Fig. 5. Accuracy comparison of proposed scheme and other approaches

4.3. Execution Time for Customer Registration in Blockchain

In TruChit the customers with satisfactory credit recommendations, register into the permissioned blockchain network. To achieve this TruChit is implemented in Hyperledger Fabric and registered 300 customers to the network. Fig. 6 depicts the execution time for adding the customers to blockchain network. This includes transaction endorsement, validation, ordering, and commitment time for registering customers to blockchain network. The results indicate that the execution time of 10 registrations is 9.15 ms and increases linearly with the number of customer registrations.

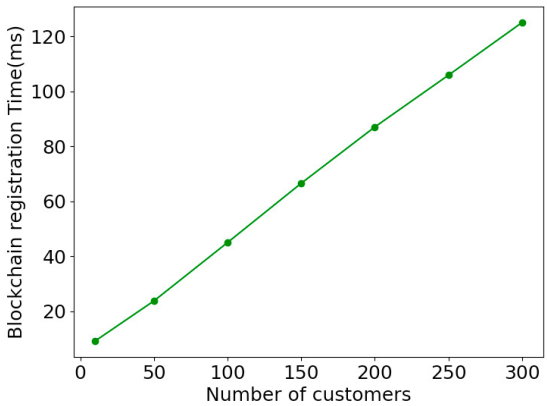

Fig. 6. Execution time for customer registration

5. Conclusion

The paper proposes TruChit, an ANFIS-based creditworthiness evaluation framework integrated with blockchain technology to ensure credibility, reliability, and transparency in chit fund system. In this, ANFIS classifier analyzes the financial behavior of subscribers, computes credit scores, and classifies them as trustworthy, or non-trustworthy. This helps chit operators to distinguish between legitimate and unscrupulous participants, as a result mitigating payment defaults. Further, in contrast to non-transparent transaction processing and record-keeping, TruChit leverages permissioned blockchain with role-based access control mechanisms. This approach ensures trust, transparency, and tamper resistance in the chit fund system. Additionally, it allows participants to track and monitor fund activities, which mitigates the risk of fraud and creates a more sustainable ecosystem. Moreover, the effectiveness of the proposed solution is demonstrated through empirical analysis of credit score dataset that involves rigorous training on 80% of the financial behavior data, while the remaining 20% is used for testing and prognosis evaluation. As a result, TruChit achieves

an accuracy of 93.5% and outperforms existing approaches, including Decision Trees, Neural Networks, SVM, Naive Bayes, and KNN classifiers. Finally, the deployment of the embedded framework in Hyperleder Fabric with 300 participants' registration demonstrates the feasibility of the proposed system. In the future, further enhancements to the creditworthiness evaluation process can be explored by using deep learning techniques and incorporating multiple alternative datasets to improve the accuracy of credit score assessments.

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